MATH 2211, SPRING 2026: HOMEWORK 6

(1) Suppose that we have a matrix $A \in M_{n\times m}(F)$ associated with a linear transformation $T : F^m \to F^n$. How would you use row reduction to verify the following:
 (a) If T is injective?
 (b) If T is surjective?
 (c) (when $m = n$) If T is an invertible linear transformation? In this case, also give a method for finding the inverse matrix A^{-1} (see Lecture 17).

(2) Show that the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 3 & -1 & 0 \\ 1 & 0 & 2 \end{bmatrix} \in M_{n\times n}(\mathbb{Q})$$

is invertible and find its inverse. Use the inverse to solve the equations

$$Ax = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \; ; \; Ax = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

(3) Consider the function

$$\begin{aligned} T : \mathrm{Poly}_{\leq 3}(F) &\longrightarrow \mathrm{Poly}_{\leq 3}(F) \\ f(x) &\mapsto f(0)x^3 + f(1)x^2 + f(1)x + f(0) \end{aligned}$$

Show that T is a linear transformation and compute its matrix with respect to the bases

$$\mathcal{B} = (1, x, x^2, x^3) \; ; \; \mathcal{B}' = (1, (x-1), (x-1)^2, (x-1)^3).$$

(4) Suppose that V and W are finite dimensional F-vector spaces, and define

$$\mathrm{Hom}_F(V, W) = \{F\text{-linear transformations } T : V \to W\}$$

(the Hom is short for homomorphism).
 (a) Show that $\mathrm{Hom}_F(V, W)$ is an F-vector space with linear combinations given by:

$$a_1T_1 + a_2T_2 : v \mapsto a_1T_1(v) + a_2T_2(v) \in W$$

 for $a_1, a_2 \in F$ and $T_1, T_2 \in \mathrm{Hom}_F(V, W)$.
 (b) If $\mathcal{B}_1$ and $\mathcal{B}_2$ are bases for V and W, respectively, with $n = \dim V$ and $m = \dim W$ show that the function

$$\begin{aligned} \alpha : \mathrm{Hom}_F(V, W) &\to M_{n\times m}(F) \\ T &\mapsto [T]_{\mathcal{B}_1, \mathcal{B}_2} \end{aligned}$$

 is an isomorphism of F-vector spaces.
 (c) What is the dimension of $\mathrm{Hom}_F(V, W)$?

An $n \times n$-matrix is **upper triangular** if it doesn't have any non-zero entries below the diagonal. That is, if $a_{i,j} = 0$ whenever $i > j$. An F-linear transformation $T : V \to V$ on a finite dimensional vector space V is **upper triangulizable** if there exists a basis $\mathcal{B}$ for V such that $[T]_{\mathcal{B}}$ is upper triangular.

(5) Show that the following are equivalent for $T : V \to V$
 (a) T is upper triangulizable.

(b) There exists a basis $\{v_1, v_2, \ldots, v_n\}$ of V such that, for each j,
$$T(v_j) = \sum_{i \leq j} b_{i,j} v_i,$$
for some $b_{i,j} \in F$.

(c) There exists a basis $\{v_1, v_2, \ldots, v_n\}$ of V such that, for each i,
$$T(\mathrm{span}(v_1, \ldots, v_i)) \leq \mathrm{span}(v_1, \ldots, v_i).$$